



15.2 Classification of polymers

1. Which of the following is regenerated fibre?
[MH CET 2011]
(A) Nylon (B) Nylon 6
(C) Rayon (D) Terylene
2. Which one is classified as a condensation polymer?
[JEE (Main) 2014]
(A) Dacron (B) Neoprene
(C) Teflon (D) Acrylonitrile
3. This is a thermoplastic:
[BCECE 2014]
(A) PVC (B) Bakelite
(C) Melamine (D) Protein
4. CORRECT statement for thermoplastic polymer is _____.
[MHT CET 2016]
(A) It does not become soft on heating under pressure
(B) It cannot be remoulded
(C) It is either linear or branched chain polymer
(D) It is cross-linked polymer

5. Which polymer among the following polymers does NOT soften on heating?

- (A) Bakelite (B) Polythene
(C) Polystyrene (D) PVC

[MH CET 2015]



15.3 Some important polymers

6. Natural rubber has _____. [NEET P-I 2016]

- (A) alternate cis- and trans-configuration
(B) random cis- and trans-configuration
(C) all cis-configuration
(D) all trans-configuration

7. cis-1,4-polyisoprene is called _____. [KCET 2019]

- (A) buna-S (B) natural rubber
(C) buna-N (D) neoprene

8. The following is used for making natural rubber hard: [BCECE 2014]

- (A) Petrol (B) H_2SO_4
(C) P_4 (D) S

9. Industrial production of polythene is done by polymerization of _____. [BCECE 2014]

- (A) methane (B) acetylene
(C) ethylene (D) styrene

10. Which of the following statements about low density polythene is FALSE? [JEE (Main) 2016]

- (A) Its synthesis requires high pressure.
(B) It is a poor conductor of electricity.
(C) Its synthesis requires dioxygen or a peroxide initiator as a catalyst.
(D) It is used in the manufacture of buckets, dust-bins, etc.

11. Ziegler-Natta catalyst is used to prepare: [KCET 2018]

- (A) Low density polythene
(B) Teflon
(C) High density polythene
(D) Nylon 6

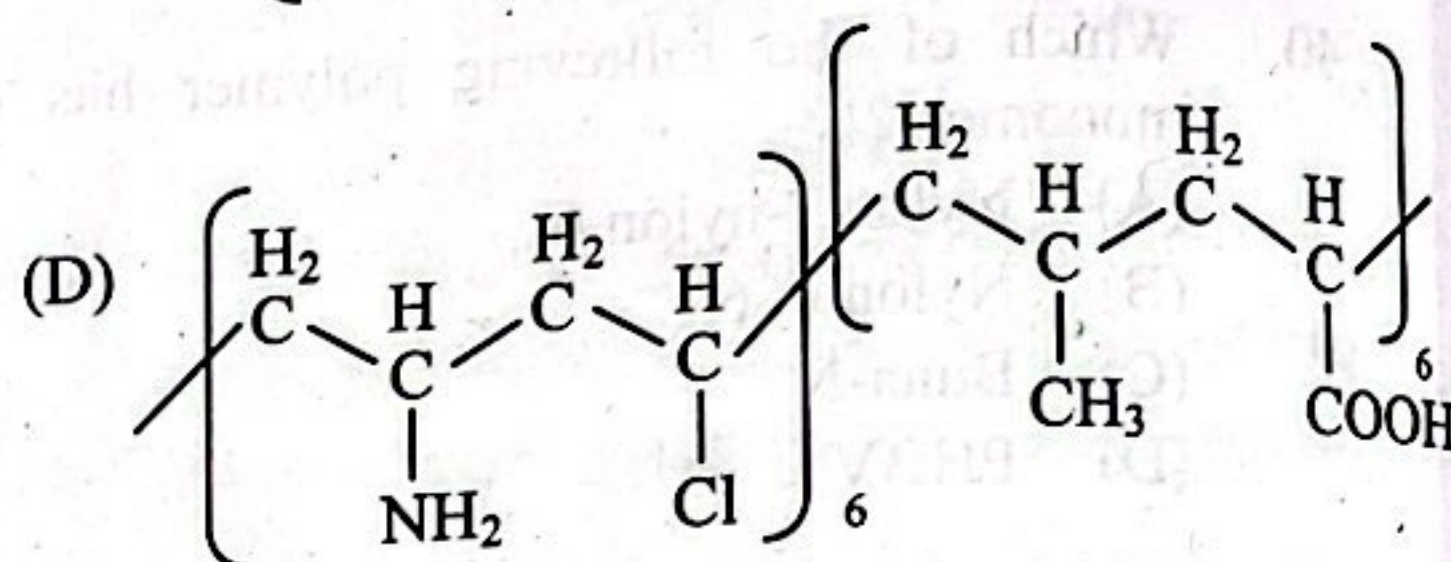
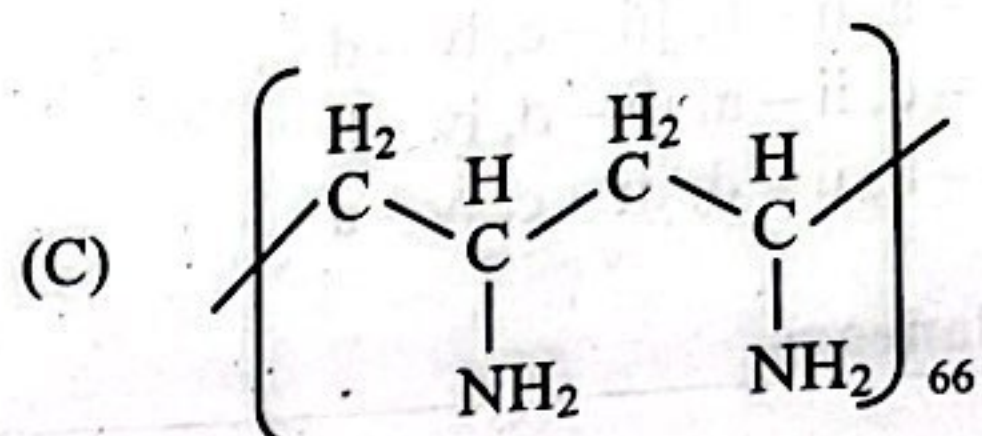
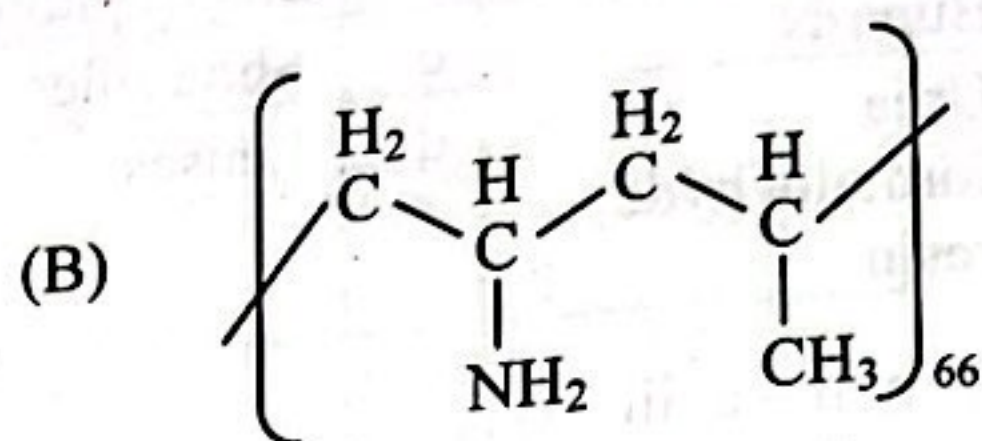
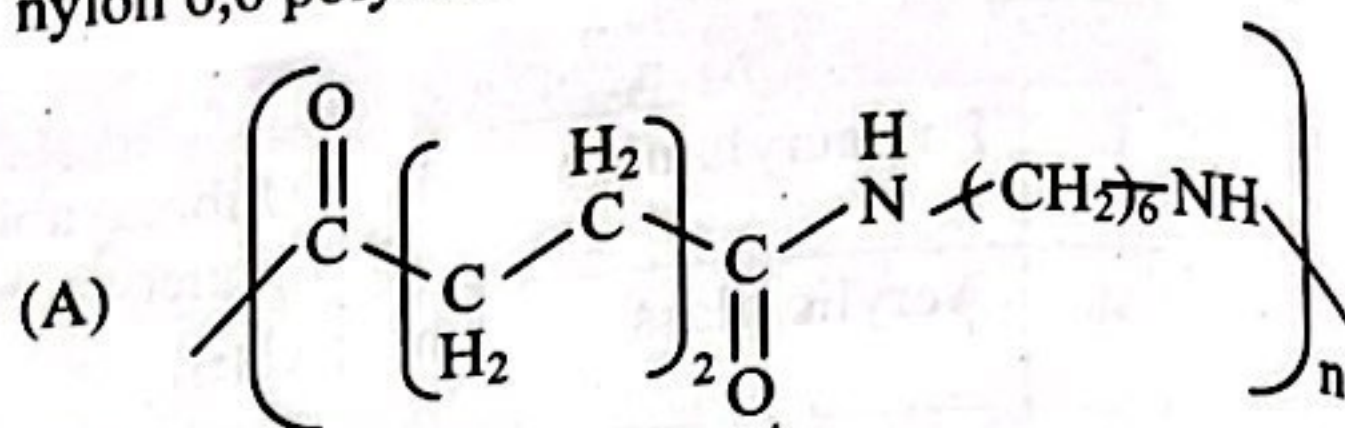
12. Orlon has monomeric unit _____. [KCET 2015]

- (A) acrolein (B) glycol
(C) vinyl cyanide (D) isoprene

13. The polymer that is used as a substitute for wool in making commercial fibres is _____. [NEET (Odisha) 2019]

- (A) buna-N
(B) melamine
(C) nylon 6,6
(D) polyacrylonitrile

14. Which one of the following structures represents nylon 6,6 polymer? [NEET P-II 2016]



15. Which one of the following is a polyamide polymer? [KCET 2018]

- (A) Terylene (B) nylon 6,6
(C) Buna-S (D) Bakelite

16. ϵ -Caprolactum is used for the manufacture of _____. [AIPMT Re-Test 2015]

- (A) terylene (B) nylon 6,6
(C) nylon 6 (D) teflon

17. Nylon 6 is made from _____. [BCECE 2015]

- (A) isoprene (B) adipic acid
(C) ϵ -caprolactum (D) styrene

18. The polymer containing five methylene groups in its repeating unit is _____. [KCET 2019]

- (A) dacron (B) bakelite
(C) nylon 6,6 (D) nylon 6

19. The formation of which of the following polymers involves hydrolysis reaction? [JEE (Main) 2017]

- (A) Nylon 6 (B) Bakelite
(C) Nylon 6,6 (D) Terylene

20. Which of the following organic compounds polymerizes to form the polyester dacron? [AIPMT 2014]

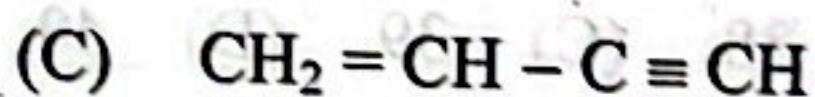
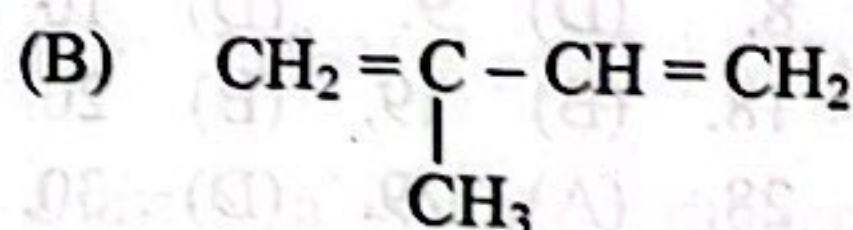
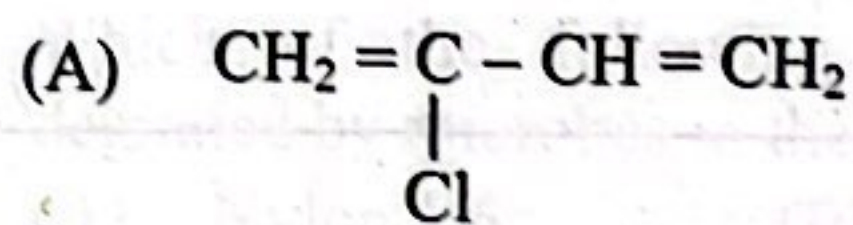
- (A) Propylene and para $\text{HO} - (\text{C}_6\text{H}_4) - \text{OH}$
(B) Benzoic acid and ethanol
(C) Terephthalic acid and ethylene glycol
(D) Benzoic acid and para $\text{HO} - (\text{C}_6\text{H}_4) - \text{OH}$

21. Terylene is NOT a _____. [KCET 2014]
 (A) copolymer
 (B) polyester fibre
 (C) chain growth polymer
 (D) step growth polymer

22. The monomers of Buna-S rubber are _____. [AIIMS 2010]
 (A) vinyl chloride and sulphur
 (B) butadiene
 (C) styrene and butadiene
 (D) isoprene and butadiene

23. In Buna-S, S stands for _____. [BCECE 2015]
 (A) sulphur (B) soft
 (C) styrene (D) sodium

24. Which is a monomer of neoprene in the following? [NEET 2013]



15.5 Biodegradable polymers

25. Which of the following is biodegradable polymer? [MH CET 2013]

- (A) Polyhydroxy butyrate-co-hydroxy valerate
 (B) Nylon 6
 (C) Bakelite
 (D) Polythene

26. Which of the following pair of monomers are used in preparation of PHBV? [GUJ CET 2014]

- (A) β -Hydroxybutyric acid, β -hydroxyvaleric acid
 (B) β -Hydroxyvaleric acid, Aminocaproic acid
 (C) β -Hydroxybutyric acid, Adipic acid
 (D) Lactic acid, Adipic acid

27. Biodegradable polymer which can be produced from glycine and aminocaproic acid is _____. [AIPMT 2015]

- (A) nylon 2-nylon 6 (B) PHBV
 (C) Buna-N (D) nylon 6, 6

28. The biodegradable polymer is _____. [NEET (UG) 2019]

- (A) nylon 2-nylon 6 (B) nylon 6
 (C) Buna-S (D) nylon 6, 6

29. Which of the following is biodegradable polymer? [MHT CET 2019]

- (A) Nylon 6 (B) Nylon 2-nylon 6
 (C) Terylene (D) Nylon 6, 6

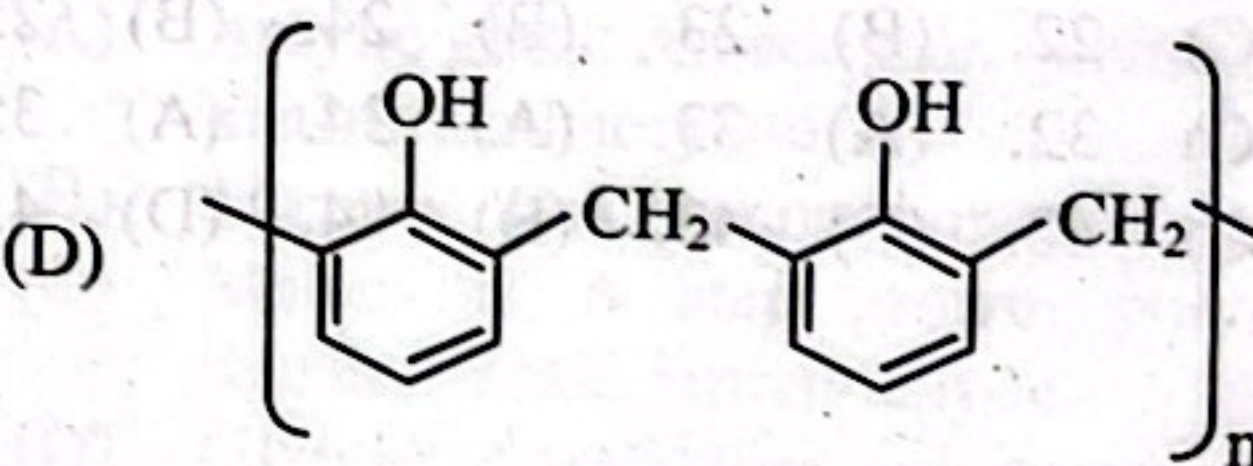
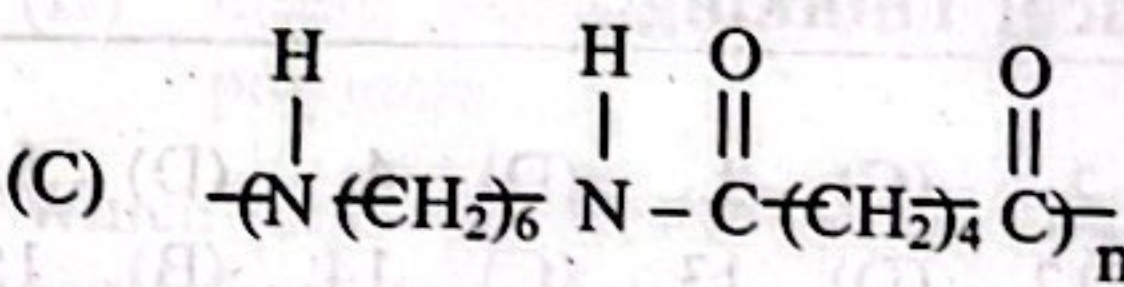
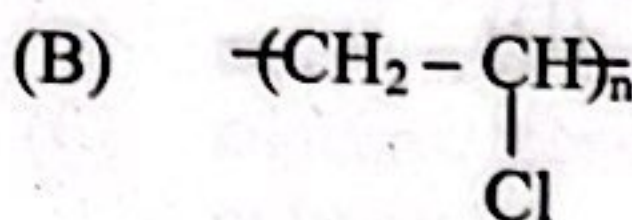
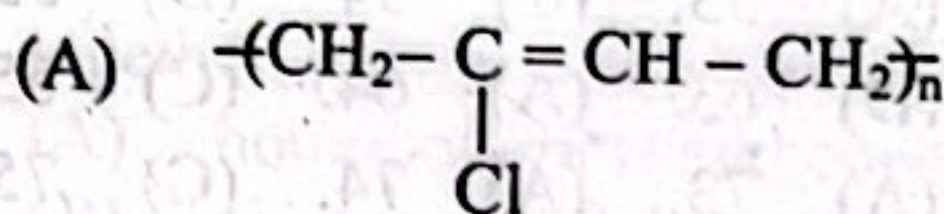
15.6 Commercially important polymers

30. Which polymer is used in the manufacture of paints and lacquers? [JEE (Main) 2015]

- (A) Bakelite
 (B) Glyptal
 (C) Polypropene
 (D) Polyvinyl chloride

Miscellaneous

31. Which of the following is an example of thermosetting polymer? [AIPMT 2014]



32. Identify homopolymers from the following. [AP EAMCET (Med.) 2019]

- | | | |
|---------------|---------------|-----------------|
| Nylon 6
I | Orlon
II | Bakelite
III |
| Glyptal
IV | Neoprene
V | |

- (A) II, III, IV (B) I, II, V
 (C) I, III, IV (D) II, IV, V

33. Identify the addition polymers from the following:

- | | |
|------------------------|-----------------|
| i. Terylene | ii. Polypropene |
| iii. Polyacrylonitrile | iv. Nylon 6 |
| v. Polyvinyl chloride | |

- [TS EAMCET (Med.) 2015]
 (A) ii, iii, v (B) i, ii
 (C) i, iv (D) ii, iv

34. The polymer used in making handles of cookers and frying pans is _____. [MHT CET 2017]

- (A) bakelite
 (B) nylon 2-nylon 6
 (C) orlon
 (D) polyvinyl chloride

35. Which of the following sets contain only non-biodegradable polymers?

- (A) Nylon 6, PVC
- (B) Nylon 6, PHBV
- (C) PHBV, Nylon 2-nylon 6
- (D) PVC, Nylon 2-nylon 6

36. Identify the CORRECT statements.

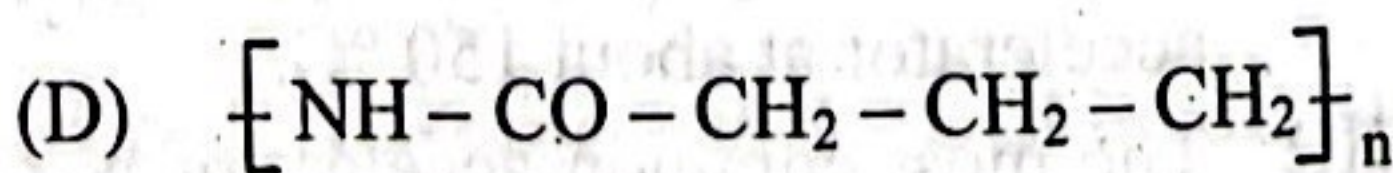
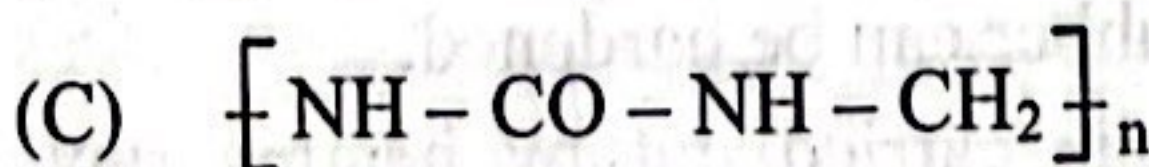
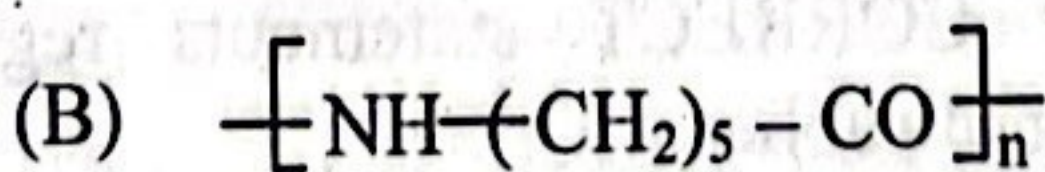
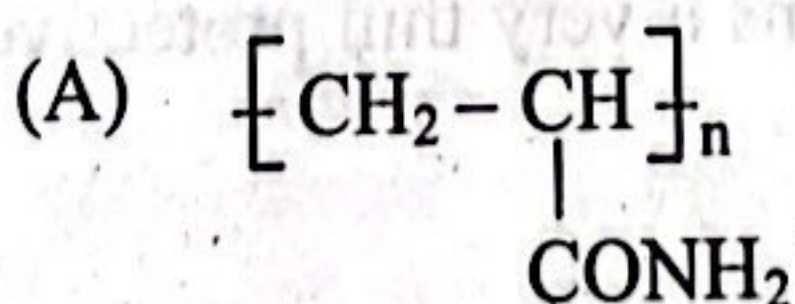
- (I) PHBV is an aliphatic polyester.
- (II) Nylon 2-nylon 6 is a copolymer of two amino acids: glycine and ϵ -amino caproic acid.
- (III) Nylon 2-nylon 6 is a polyamide.
- (IV) PHBV is a copolymer of two bifunctional α -hydroxy carboxylic acids.

- (A) (I) and (II)
- (B) (II) and (IV)
- (C) (I), (II), (III)
- (D) (I), (II), (III), (IV)



15.6 Commercially important polymers

37. The structure of urea formaldehyde polymer is _____.



38. Which of the following fibre is used to make electrical and communication hardware?

- | | |
|-------------------|-------------|
| (A) Glyptal | (B) Perspex |
| (C) Polycarbonate | (D) Buna-S |

39. Match the polymers in column I with its use in column II

	Polymers		Uses
i.	Polyacrylamide	a.	Unbreakable dinnerwares
ii.	Acrylic glass	b.	Used in electrophoresis
iii.	Buna-N	c.	Shoe soles
iv.	Urea formaldehyde resin	d.	Lenses

- (A) i - c, ii - a, iii - b iv - d
 (B) i - a, ii - b, iii - c, iv - d
 (C) i - c, ii - a, iii - d, iv - b
 (D) i - b, ii - d, iii - c, iv - a



Miscellaneous

40. Which of the following polymer has chiral monomers?
 (A) Nylon 2-nylon 6
 (B) Nylon 6, 6
 (C) Buna-N
 (D) PHBV
41. Which of the following is a naturally occurring substance but NOT a polymer?
 (A) DNA (B) Cellulose
 (C) ATP (D) Protein
42. Letter 'N' in Buna-N stands for _____.
 (A) normal (B) nitrogen
 (C) nitrile (D) sodium
43. Which one of the following pairs is NOT correctly matched?
 (A) Terylene—condensation polymer of terephthalic acid and ethylene glycol.
 (B) Teflon—thermally stable cross linked polymer of phenol and formaldehyde.
 (C) Urea formaldehyde resin—A cross linked polymer of urea and formaldehyde.
 (D) Buna-S—a copolymer of butadiene and styrene.
44. Which of the following is chain growth polymer?
 (A) Neoprene (B) Buna-S
 (C) PAN (D) All of these
45. Which of the following polymer is formed from a monomer that does not show geometric isomerism?
 (A) Polyisoprene
 (B) Polytetrafluoroethene
 (C) Neoprene
 (D) PVC